

Python String and Tuple Operations

Objective:

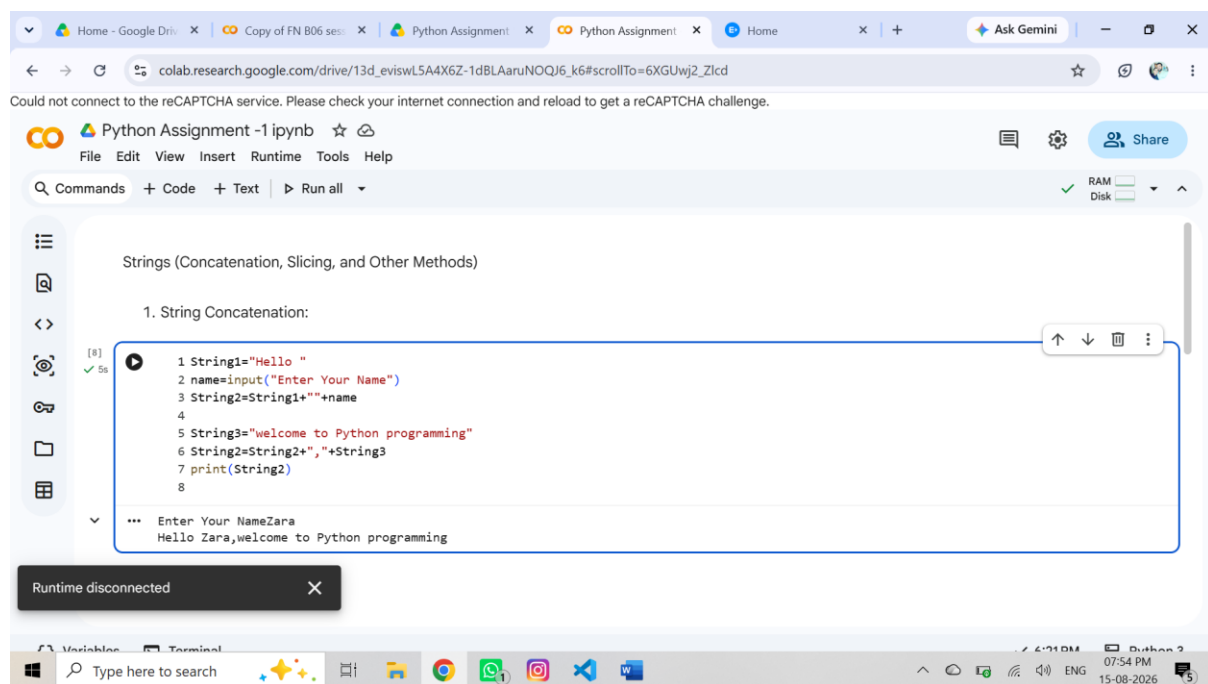
The objective of this assignment is to understand and practice the basic concepts of Python programming, especially string concatenation, indexing, slicing, string methods, and tuple operations. It helps to learn how to create, access, modify, combine, and manipulate strings and tuples using Python. The assignment also aims to improve logical thinking, coding skills, and understanding of Python's built-in functions and operations.

1. String Concatenation:

Write a Python program that takes two strings, i.e., string 1 "Hello" and string 2 "get name" as input from the user, and concatenates them together. Display the concatenated string as the output.

Sample Output:

Hello Zara, welcome to Python programming



The screenshot shows a Google Colab notebook titled "Python Assignment -1 ipynb". The code cell contains the following Python code:

```
1 String1="Hello "
2 name=input("Enter Your Name")
3 String2=String1+" "+name
4
5 String3="welcome to Python programming"
6 String2=String2+","+String3
7 print(String2)
8
```

The output of the code is:

```
Enter Your NameZara
Hello Zara,welcome to Python programming
```

A "Runtime disconnected" message is visible at the bottom of the code cell.

2. String Slicing and Indexing:

Write a Python program using the above concatenated string as input and perform the following tasks:

- Print the first character of the string.
- Print the last character of the string.
- Print the first 5 characters of the string.
- Print the last 11 characters of the string.
- Print the string in reverse.
- Use slicing and print the word "Python" from the existing string.

The image displays two screenshots of a Google Colab notebook titled "Python Assignment -1 ipynb". The notebook is open to a section titled "2. String Slicing and Indexing".

Top Screenshot: Shows the initial code and its output.

```
[10] 1 #existing concatenated string
      2 text="Hello Zara,welcome to Python programming"
      3
```

Output: [15] 1 #a.First character
2 print("a.First character:",text[0])
a.First character: H

Output: [16] 1 #b.last character
2 print("b.last character:",text[-1])
b.last character: g

Output: [17] 1 #c.firts 5 characters

Bottom Screenshot: Shows the continuation of the code and its output.

```
[17] 1 #c.firts 5 characters
      2 print("c.firts 5 characters:",text[:6])
```

Output: c.firts 5 characters: Hello

```
[25] 1 #d.last 11 characters
      2 print("d.last 11 characters:",text[-11:])
```

Output: d.last 11 characters: programming

```
[26] 1 #e.reverse the string
      2 print("e.reverse the string:",text[::-1])
```

Output: e.reverse the string: gnimmarginorp nohtyP ot emoclew,araZ olleH

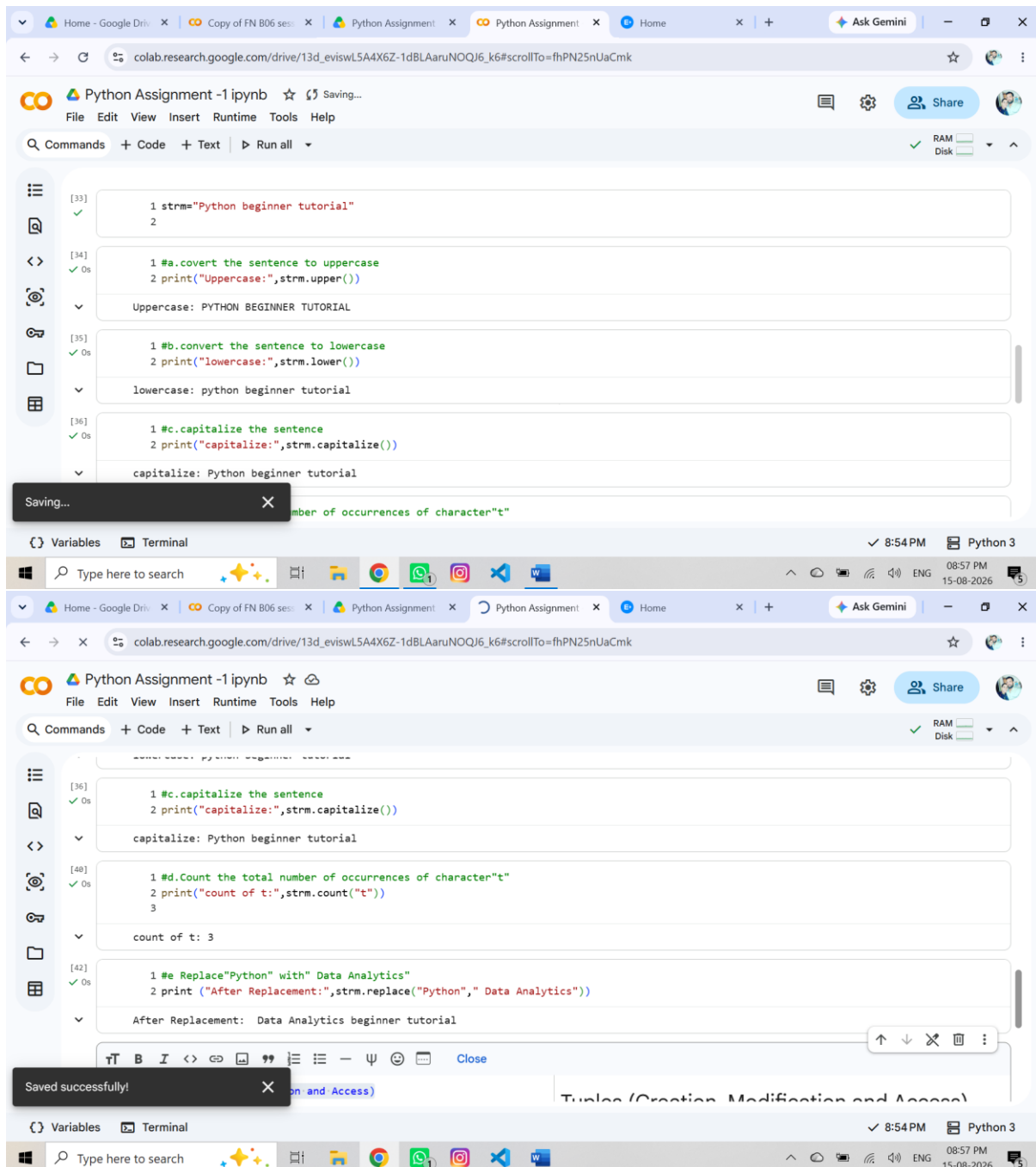
```
[32] 1 #f.Print the word "Python"
      2 print("f word Python:",text[22:28])
```

Output: f word Python: Python

3. String Methods:

Write a Python program that takes a string, strm= "Python beginner tutorial," and performs the following tasks:

- Convert the sentence to uppercase.
- Convert the sentence to lowercase.
- Use Capitalize and return the sentence to the original input form.
- Count the total number of occurrences of character 't' in the string.
- Replace all occurrences of "Python" with "Data Analytics" in the input string strm = "Python beginner tutorial."



Tuples (Creation, Modification and Access) :

Create the 1st tuple with values -> (10, 20, 30) and the 2nd tuple with values -> (40, 50, 60):

- Concatenate the two tuples and store it in "t_combine"
- Repeat the elements of "t_combine" 3 times
- Access the 3rd element from "t_combine"
- Access the first three elements from "t_combine"
- Access the last three elements from "t_combine"

```
[43]
1 #Tuples- Creation, Modification and Access
2 #Create two tuples
3 t1=(10,20,30)
4 t2=(40,50,60)

[44]
1 #a.concatenate the two tuples
2 t_combine=t1+t2
3 print("Combined Tuple:",t_combine)

Combined Tuple: (10, 20, 30, 40, 50, 60)

[45]
1 #b.Repeat the element of t_combine 3 times
2 print("Repeated Tuple:",t_combine*3)

Repeated Tuple: (10, 20, 30, 40, 50, 60, 10, 20, 30, 40, 50, 60, 10, 20, 30, 40, 50, 60)
```

```
[46]
1 #c.Access the 3rd element from t_combine
2 print("3rd element:",t_combine[2])
3

3rd element: 30

[53]
1 #d Access the first three elements
2 print("First 3 elements:",t_combine[:3])

... First 3 elements: (10, 20, 30)

[54]
1 #e. last three elements
2 print("last three elements:",t_combine[-3:])

last three elements: (40, 50, 60)
```

Analysis

The analysis of this assignment focuses on understanding how Python handles strings and tuples. String operations such as concatenation, indexing, slicing, and built-in methods are used to manipulate and analyze text. Tuple operations such as concatenation, repetition, and slicing are used to store and access multiple values efficiently. These programs help in understanding Python syntax, data handling, and basic programming logic.

Conclusion:

This assignment helped me understand the basic operations of Python strings and tuples. I learned how to concatenate, slice, index, and modify strings using built-in methods. I also learned how to create and manipulate tuples using different operations. These concepts are useful for developing a strong foundation in Python programming.